

MASHRUBA TASNEM OISHI

Aspiring University Lecturer | AI & Cybersecurity Researcher | Full-Stack Developer

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📄 PROFESSIONAL SUMMARY

Computer Science & Engineering graduate from IUBAT (CGPA: 3.60/4.00) with expertise in Explainable AI (XAI), Machine Learning, Cybersecurity, and Full-Stack Web Development. Conducted undergraduate research on a leakage-free and explainable network intrusion detection framework using SHAP, Random Forest, and XGBoost, with the manuscript currently being prepared for journal submission. Experienced in developing AI-integrated full-stack web applications using PHP, MySQL, JavaScript, and Bootstrap. Passionate about teaching core Computer Science courses, mentoring students, and advancing research in Trustworthy AI and Cybersecurity.

🎓 EDUCATION

International University of Business Agriculture & Technology (IUBAT),

B.Sc. in Computer Science & Engineering (BCSE)

06/2026 | Dhaka, Bangladesh

- CGPA: 3.60/4.00

RAJUK Uttara Model College, Dhaka, HSC

2019 | Dhaka, Bangladesh

GPA: 5.00/5.00

RAJUK Uttara Model College, SSC

2017 | Dhaka, Bangladesh

GPA: 5.00/5.00

🧩 TEACHING INTERESTS

- Programming Fundamentals, Object-Oriented Programming, Data Structures & Algorithms, Database Management Systems, Computer Networks, Software Engineering, Management Information Systems, System Analysis & Design, Digital Logic Design.
- Open to teaching other courses as needed.

🧠 TECHNICAL SKILLS

Languages – Python, PHP, Java, C#, C

Machine Learning – Scikit-learn, Pandas, NumPy, XGBoost, Random Forest, SHAP

Web & Database – HTML/CSS, Bootstrap, JavaScript, jQuery, MySQL, SQL Server

Tools – Git, GitHub, VS Code, Visual Studio, Google Colab, Kaggle

🌐 LANGUAGES

Bengali

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English

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📁 PROFESSIONAL EXPERIENCE

Web Developer Intern (Academic Internship), SoftDeft

02/2026 – 05/2026 | Gazipur

- Designed and built a full-stack Animal Rescue & Sanctuary Management System (PHP, MySQL, Bootstrap, JavaScript) with 4 role-based portals, 20+ relational database tables, secure CRUD operations, and SSLCommerz payment integration.
- Integrated AI-assisted features – Groq-powered SOS triage, chatbot, and OpenStreetMap-based vet clinic finder – to improve rescue response and accessibility.

📄 UNDERGRADUATE THESIS & PUBLICATION

XAI-NIDS: A Leakage-Free and Explainable Machine Learning Framework for Trustworthy Network Intrusion Detection

Supervisor: Md. Alamin Talukder (Stanford–Elsevier Top 2% Scientist) | Manuscript in preparation, targeting a Springer venue (Int'l Journal of Machine Learning and Cybernetics)

- Built a leakage-free ML framework for intrusion detection on the UNSW-NB15 dataset; benchmarked Random Forest and XGBoost classifiers.
- Applied SHAP-based Explainable AI to improve model transparency; final XGBoost model reached 90.52% accuracy and 90.43% F1-score.

🧩 RESEARCH INTERESTS

Artificial Intelligence, Explainable AI (XAI), Machine Learning, Trustworthy AI, Cybersecurity, Network Intrusion Detection, Computer Vision, Data Science

📁 ACADEMIC PROJECTS

Explainable Network Intrusion Detection System, Python, Scikit-learn, XGBoost, SHAP; leakage-free IDS on UNSW-NB15.

Animal Rescue & Sanctuary Management System, PHP, MySQL, Bootstrap, JavaScript; rescue, adoption, and donation management.

Home Automation System,, C, Arduino, Sensors; smart home automation using sensor-based device control.

🔗 REFERENCES

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Technology (IUBAT)

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